PROJECT REVIEW

Network Connectivity

Using assets/email_network.txt

An academic review of graph construction, connectivity analysis, and network visualization with Python and NetworkX.



Data Science • Graph Theory • Network Analysis

Project Overview

What the project studies

Dataset

assets/email_network.txt

A text file containing email-network relationship data.

Representation

Participants are modeled as nodes; recorded relationships are modeled as edges.

Tools

Python and NetworkX support graph loading, connectivity checks, and visualization.

Core question: Are all participants connected through paths, or does the network contain separate groups?

Problem Statement & Motivation

Why network connectivity matters

Email datasets can contain many relationships that are difficult to inspect manually.

Graph analysis provides a structured way to study communication links among participants.

Connectivity analysis can reveal separate groups, isolated nodes, and the reachability of participants.

The analysis describes recorded connections; it does not automatically measure relationship strength or communication quality.

Project Objectives

Specific goals of the analysis

Load and inspect the email-network text file.

Represent participants and relationships as a graph.

Calculate the numbers of nodes and edges.

Determine graph connectivity and identify connected components.

Count isolated nodes and visualize the network structure.

Interpret results while acknowledging data and modeling limitations.

Methodology

A reproducible analytical workflow

1

Inspect

Check file format, delimiters, and record structure

2

Load

Read edge data into NetworkX

3

Analyze

Compute graph size and connectivity measures

4

Visualize

Plot nodes and edges to inspect structure

5

Interpret

Explain findings and document limitations

Important: choose the graph type and file-reading method only after inspecting the actual data.

Key Concepts in Graph Theory

How to interpret the network

Node

A participant or entity represented in the network.

Edge

A recorded relationship between two nodes.

Connected Component

A group in which every node can reach every other node by a path.

Isolated Node

A node with no edges connecting it to other nodes.

Python Implementation

Example for a whitespace-delimited edge list

```
import networkx as nx

file_path = "assets/email_network.txt"
G = nx.read_edgelist(
    file_path,
    nodetype=str,
    create_using=nx.Graph()
)

print("Nodes:", G.number_of_nodes())
print("Edges:", G.number_of_edges())
print("Connected:", nx.is_connected(G) if G else False)
print("Components:", nx.number_connected_components(G))
print("Isolates:", len(list(nx.isolates(G))))
```

Assumption

Each line contains two node identifiers separated by whitespace.

Validation

If the file uses another format, adjust the parser and graph type.

Metrics & Their Meaning

Quantitative measures to report

Node count

Number of distinct participants

Edge count

Number of recorded links

Component count

Number of separate connected groups

Largest component

Size of the biggest connected group

Isolated-node count

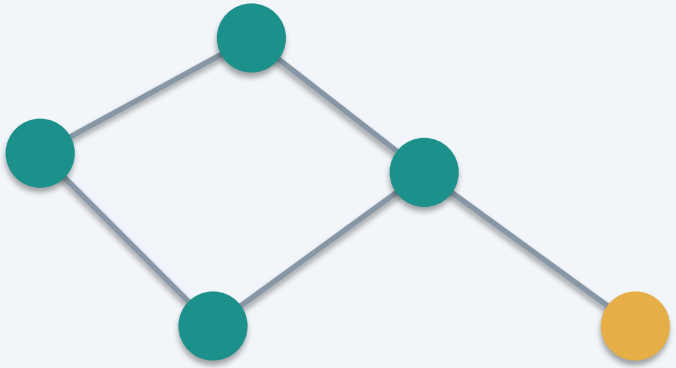
Participants with degree zero

Connectivity status

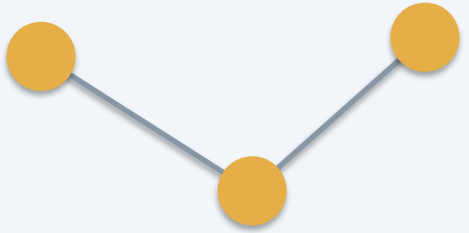
Whether the entire undirected graph is connected

Network Visualization

A visual companion to the numerical analysis



Conceptual illustration only



Clusters may suggest groups, but only paths and edges establish connectivity.

Limitations & Future Work

How the analysis could be extended

Limitations

- File format must be verified
- Direction of emails may matter
- Static graphs hide changes over time
- Missing data can distort relationships
- Visual layouts can become crowded

Future Improvements

- Use a directed graph when appropriate
- Add weighted edges for message volume
- Analyze degree and betweenness centrality
- Detect communities or clusters
- Compare network structure over time

CONCLUSION

From email records to network insight

NetworkX converts relationship records into a graph that can be measured and visualized.

Connectivity metrics reveal whether the network forms one connected structure or separate groups.

Reliable conclusions require checking the dataset format and reporting actual computed results.

Thank you • Questions

References

Foundational sources

Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). Exploring Network Structure, Dynamics, and Function using NetworkX. Proceedings of the 7th Python in Science Conference.

NetworkX Developers. NetworkX Documentation. https://networkx.org/documentation/stable/

Newman, M. (2010). Networks: An Introduction. Oxford University Press.